UBS Global I/O Semiconductors

SEMICON Taiwan 2026: 5 key takeaways for advanced packaging, advanced logic, CPO & test

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An event at record scale

SEMICON Taiwan 2026, held on 31 Aug - 4 Sept, was one of the largest electronics shows in Taiwan in recent years, reflecting increased attention on the importance of the semiconductor industry and supply chain in Taiwan. According to SEMICON's official press release, the event participation was at an unprecedented scale attracting over 1,300 exhibitors and 4,300 booths (vs 1,200 companies and 4,100 booths in 2025), and over 100,000 visitors. Key industry themes the event focused on were silicon photonics, CPO, heterogeneous integration, 3DIC, testing, quantum computing, and AI power semis. See our takeaways from the major technology forums on pages 3-20.

What are our key findings?

1) For **advanced packaging**, our sense is that equipment suppliers are becoming more constructive about TSMC's panel level packaging CoPoS supported by the meaningful resources invested by TSMC and the whole supply chain. See our recent deep dive on CoPoS & EMIB. We believe TSMC's goal is to finalize 310x310mm² process and equipment selections by mid-2027 to mass produce in 2028. With the incremental progress made on **CoPoS**, TSMC appears to become slightly more conservative about its own CoWoS expansion into H227 and may lean more toward ASE for CoW outsourcing. We raise **CoWoS** industry capacity by end-2027 to 260kwpm from 250kwpm, up from 160kwpm end-2026. We keep TSMC's capacity at 180kwpm end-2027 but lift our ASE estimate to 60kwpm (from 50kwpm) and keep Amkor at 20kwpm. **Stronger ASE expansion** should benefit GPTC, the main wet-process tool supplier, and we expect its tool shipment to reach 400+ sets in 2027, up from prior estimate of 220 and 300-350 in 2026/27E. **2)** On **Intel's EMIB-T**, industry feedback is positive regarding the yield improvement and overall execution, consistent with what we highlighted in July. We are constructive on EMIB-T's supply supporting >2m substrate volume for MediaTek's TPU v9 demand in 2028. Thermal compression bonder (TCB) demand should pick up strongly for silicon bridge integration in substrates. We believe Toray in Japan is the current leading supplier. ASMPT is developing the tool for qualification in early 2027. **3)** On **front-end logic**, TSMC's expansion continues to accelerate in 2027 and is now tracking toward our estimates of ~190kwpm for N3 and 140kwpm for N2 by end-2027. A14 R&D is progressing smoothly for mass production in H228. Meanwhile, Intel's 18A is also ramping steadily for internal CPU products. **4) Co-packaged optics (CPO)** remains a key focus. The industry is progressing toward mass production for scale-out from now through 2027, hybrid scale up in 2028-29, and full CPO scale up in 2029-30, in our view. TSMC showcased its COUPE roadmap to deliver 200G/lane in 2026 and scale to 400G/lane by 2030. **5) Test** is becoming increasingly critical to support **heterogeneous integration. Probing (wafer level testing)** is seeing significant content upside by evaluating chip performance earlier, which reduces overall supply chain cost. TSMC's probing demand should sustain strong growth and benefit ASE/KYEC as outsourcing partners. Specifically for **CPO testing**, the trend may be to streamline insertion 2 (EIC on PIC wafer) given technology complexity, and to lean more toward insertion 3 (singulated optical engine). FiconTEC, the leading test handler supplier for CPO, expects to ship 1k systems in next 12-18 months, outpacing 2k shipment in the last 25 years.

Stock picks

For Taiwan semis, our top picks are MediaTek (Key Call Buy), TSMC, UMC, Win Semi, ASE, and Aspeed. The robust industry expansion across leading edge logic, advanced packaging and CPO should benefit equipment suppliers including All Ring, GPTC, Chroma, Hon Precision, and Gudeng.

This report has been prepared by UBS Securities Pte. Ltd., Taipei Branch. **ANALYST CERTIFICATION AND REQUIRED DISCLOSURES, INCLUDING INFORMATION ON THE QUANTITATIVE RESEARCH REVIEW PUBLISHED BY UBS, BEGIN ON PAGE 22.**
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Silicon Photonics Global Summit

"Scaling silicon photonics and NPO/CPO test for high-volume manufacturing"
Speaker: Mr. Aether Lee, Advantest Europe GmbH

- **Multi-stage HVM test flow and opto-electronic probe innovation.** Advantest framed testing throughput as the primary bottleneck across the silicon photonics (SiPh) supply chain as market demand transitions from scale-out transceivers to scale-up XPU fabrics, and future optical compute/memory disaggregation. To support insertion 1 (PIC wafer probe), Advantest integrated its V93000 platform into the Triton test cell alongside FormFactor's CM300SiPh probe card and TEL Prexa probers. Advantest has been spending decent R&D efforts in pushing opto-electronic wafer level probe card, from its proof of concept (PoC) release for passive alignment with Jenoptik in 2024, to high precision active alignment POC for grating and edge couplers with Technoprobe released in 2025-26.

- **Double sided wafer/OE probing for heterogeneous engines.** Addressing next generation architectures like TSMC COUPE that require simultaneous top-and-bottom probing, Advantest is developing double-sided electro-optical probe solutions to secure Known Good Optical Engines prior to advanced packaging. For insertion 2 (EIC-PIC bonded wafers), the test setup deploys top side electrical probes and bottom side FAU to overcome wafer warpage and fiber alignment constraints, with HVM qualification targeted for mid-2027. For insertion 3 (singulated OE dies), Advantest is collaborating with MPI (utilizing the DTS650-DI platform) to resolve FAU indexing and high accuracy active alignment challenges, targeting HVM qualification by early 2027.

- **NPO/CPO package final test and instrumentation scalability.** For insertion 4 and 5 (socket based final test and system level test), testing architectures are splitting into optical loopback with efficient end-to-end TX/RX validation and electrical built-in self-test (eBIST) to suppress overall cost of test (CoT). In the absence of standardized detachable optical connectors, Advantest is enabling both passive and actuator-driven active connector handling via integrated optical load boards. Crucially, legacy rack & stack instrumentation and its non-scalability with fiber count becomes the bottleneck, as it cannot scale alongside IO bandwidth and DUT fiber. Advantest offers its V93000 SmarTest Optical Control Library that manages tight integration and high density ATE optical instrumentation.

"Advanced packaging for elevating AI system performance"
Speaker: Dr. C.P. Hung, ASE

- **Heterogeneous integration roadmap with reticle scaling.** ASE projects package reticle sizes expanding aggressively to accommodate next-generation compute and memory, scaling from 5.5x reticle with HBM4 to 7.5x with HBM4E by 2027 and reaching 10x reticle with HBM5 by 2030. To overcome interposer routing density constraints, ASE is transitioning from standard 2.5D Si interposers and Fan-Out Chip on Substrate (FOCoS) to FOCoS-Bridge (FOCoS-B), embedding bridge dies with fine L/S to deliver up to 200x I/O density. To tackle throughput and cost bottlenecks of massive 7.5x reticle designs—which yield only 6 interposers on a standard 300mm wafer—ASE is advancing panel-level packaging (PLP), achieving a 1.5x throughput gain on 300x310mm panels with 9 interposers, and up to 6.0x to 8.1x throughput on 600x620mm panels with 36 to 49 interposers depending on cross line.

- **Optical engine TAM inflection and 3D EIC/PIC architecture evolution.** Optical interconnects deliver a 32x bandwidth increase and 6x energy reduction compared to traditional electrical board routing over long reaches. Driven by massive scale-up AI cluster buildouts, Industry's 2031 market forecast for optical engines is trending up substantially. In particular, scale-up optical engine revenue is forecasted to surge to US$105.9bn (up from US$548m estimate released a year ago), reflecting a 214% CAGR from 2026 to 2031. To support this demand, OE packaging is advancing rapidly from legacy CoW+WB and FCBGA towards 3D FOPOP and 3D IC.

- **CPO fiber assembly and multi-stage test infrastructure.** For CPO deployment, ASE is driving wafer-level optical coupler assembly with active

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alignment and passive alignment for edge couplers and grating couplers with minimized insertion loss. Manufacturing flow requires a rigorous four-stage test architecture, with the 4 CPO test insertions: Insertion 1, EIC/PIC wafer test. Insertion 2, CoW wafer test. Insertion 3, OE unit test. Insertion 4, CPO test. Goal is to get the Known Good OE, FAU, and CPO during assembly stages.

"AI is scaling faster than test: rethinking wafer, chip and module test for the photonics era"

Speaker: Mr. Andre Lalonde, ficonTEC

- **System level disaggregation and four stage test architectures**. ficonTEC argues that Moore's Law has transitioned into a packaging problem, necessitating an evolved test architecture that lumps GPUs/switches, HBM, EICs, PICs, lasers, and fibers. To prevent costly field failures and ensure traceability back to the wafer level, ficonTEC outlines the rigorous four-insertion manufacturing test flow: insertion 1 for single-sided PIC/EIC wafer test. Insertion 2 for dual-sided electro-optical bonded wafer testing. Insertion 3 for singulated die and optical engine testing to guarantee 100% known good die (KGD). Insertion 4, for module test and pre-system qualification.

- **Wafer and die-level probing innovations.** For insertion 1, the WTS-xx single-side wafer tester incorporates laser cleaning capabilities through a partnership with FEMTUM alongside dedicated EIC and SiPh testing. As for insertion 2, ficonTEC introduced the WTF-xx double-sided wafer tester, which is compatible with both Advantest and Teradyne ATE platforms. The insertion 2 stack – comprising a top electrical probe card, the wafer, a thermal chuck, and a bottom FAU interface—must overcome severe mechanical warpage challenges when handling stacked wafers, utilizing a permanent FAU aligned to the lens array of the tester without physical contact. At insertion 3, double-sided chip testing currently running at OSATs handles 224G per channel data rates, requiring high unit per hour (UPH) throughput, meticulous dirt mitigation, and maintaining a near 100% contact rating on high-speed sockets.

- **Module test complexity and global manufacturing scalability.** Insertion 4 module testing (targeting form factors like OSFP, ELSFP, and multi-mechanical CPO architectures) faces significant scaling bottlenecks. Managing hundreds of laser fibers, mitigating dirt contamination to maintain 100% yield, and handling multi-thermal zones for devices with varied power profiles present critical challenges for standalone engineering-supervised instruments. Highlighting unprecedented capacity demand, ficonTEC expects to ship 1,000 systems in the next 12 to 18 months, compared to 2,000 systems shipped over the past 25 years. To achieve this scale, ficonTEC is executing a "designed in Germany, scaled in Taiwan" strategy, leveraging Taiwanese partnerships for co-development, extensive local production experience, and scalable capacity.

"How COUPE redefines AI interconnects"

Speaker: Mr. Ming Fa Chen, TSMC

- **Optical I/O addresses the AI I/O bottleneck.** TSMC highlighted that AI computing demand is accelerating rapidly, with global AI token usage projected to increase 24x by 2030. While compute performance scales at roughly 3x every two years, memory bandwidth grows at 1.6x and I/O bandwidth only 1.4x, making communication a key system bottleneck. TSMC argues that optical interconnects are needed because copper transmission becomes constrained at high frequencies and longer distances

- **COUPE is TSMC's silicon photonics platform for AI systems.** TSMC's COUPE (Compact Universal Photonic Engine) uses SoIC bonding to integrate electronic ICs (EICs) and photonic ICs (PICs) into a highly integrated optical engine. The platform supports both grating-coupling and edge-coupling architectures and includes critical devices such as micro-ring modulators, grating couplers, tip couplers, and photodetectors. TSMC also provides a comprehensive silicon photonics PDK to help wide base of customers accelerate design and time-to-market.

- **CPO and optical-on-interposer deliver significant efficiency gains**. The speaker compared conventional copper interconnects with co-packaged optics (CPO) and optics-on-interposer (OOI) architectures. According to the presentation, CPO and OOI can improve power efficiency by roughly 4x-10x and reduce latency by 10x-20x versus copper-based solutions. The company emphasized that

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achieving these benefits depends on a highly integrated, low-loss optical engine.

- **TSMC continues to scale COUPE performance through ecosystem collaboration.** COUPE bandwidth scales through higher lane speeds, more lanes, and wavelength-division multiplexing (WDM), while future versions can further integrate modulators, SOAs, and light sources into the platform. The speaker stated that TSMC will continue to improve COUPE for both CWDM and DWDM applications and that integration with CoWoS will rely on System and Technology Co-Optimization (STCO) across ecosystem partners. Industry adoption is also supported by growing demand for optical transceivers and increasing commitment to silicon photonics and CPO deployment.

"Optical interconnect technology for AI / ML networks"
Speaker: Mr. Matthew Sysak, Lumentum

- **Lumentum's optical portfolio is built on InP, GaAs, and optical switching technologies.** Lumentum identified Indium Phosphide (InP) as its platform for high-performance lasers, modulators, and transceivers, Gallium Arsenide (GaAs) for high-volume laser manufacturing, and MEMS/LCOS technologies for optical switching. The company presented these three technology areas as its foundational expertise for AI optical interconnects.
- **High-power InP lasers are a key focus for CPO deployments.** Lumentum highlighted its InP technology leadership through milestones including 400G-class EMLs, a 400mW ultra-high-power (UHP) CPO laser, an 800mW super-high-power (SHP) CPO laser, and a 16-channel DWDM UHP laser platform. The company also showed NVIDIA Spectrum-X Photonics and stated the platform is enabled by Lumentum's high-reliability, ultra-high-power laser technology.
- **VCSEL (Vertical-Cavity Surface-Emitting Laser) and CMOS (Complementary Metal-Oxide-Semiconductor) integration targets future scale-up interconnects.** Lumentum demonstrated 1060nm VCSEL technology integrated with advanced CMOS, supporting more than 80 Tbps of bidirectional optical bandwidth and over 1.5 Tbps/mm bandwidth density. The company also highlighted MEMS-based optical circuit switches, including a 300×300 optical switch architecture, as a potential solution for reducing network power consumption in large-scale GPU deployments.

"Breaking through the DCI data rate barrier"
Speaker: Dr. Wenchi Ting, UMC

- **Classic bandwidth scaling won't go on forever**. The scaling trend for higher data rates will not likely go on forever, at least not in a brute force manner. Both electronics and optics face challenges at 400Gbps/channel and above. For instance, SerDes is facing CMOS performance limitations and both photo diodes and Si modulators face bandwidth limitations
- **Industry is trying to find a new modulator.** No matter if its MZM, MRM, or SiGe EAM modulators, there will be thermal and bandwidth challenges that limits its data rate expansion. That's why other candidates start getting utilised, with InP and TFLN being the most notable, promising ones, due to their matureness as compared to the other solutions (organic, graphene, etc.). With excellent electrical and optical properties, coupled with mature TFLN substrate ecosystem, TFLN is the leading candidate for 400Gbps modulators. On the other hand, InP is very expensive and is harder to integrate. Further down the road coherent technology, coherent lite, in particular, will become an interesting technology for data center interconnect.
- **UMC's capabilities in SiPh.** UMC's TFLN program started back in 2021, with 6" in mass production with good yield and now shifting to 8" with sampling in progress. Integrating TFLN with PIC will be a challenge but also an opportunity, with advanced packaging schemes and micro transfer printing both are promising solutions. UMC is developing the integration technology, leveraging expertise in PIC, TFLN, and advanced packaging.

"Foundational photonic building blocks empowering next generation AI infrastructure"
Speaker: Dr. Nicolas Harris, Lightmatter

- **Accelerating AI training and inference workloads.** Lightmatter positions

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photonic interconnects as the critical lever for unlocking single-brain compute scaling across AI training and inference, The "Passage" platform significantly reduces training time for trillion-parameter MoE models; a 512-GPU pod utilizing 32 Tbps scale-up bandwidth achieves a 0.37x time-to-train relative to a 14.4 Tbps copper baseline distributed across four 144-GPU pods. For inference, Passage lowers Time to First Token (TTFT) by up to 3x for 131k context lengths and improves inter-token latency for decode by 2x compared to copper alternatives.

- **Passage L20 universal optical engine and BiDi architecture.** The passage L20 universal near-package optics (NPO) optical engine is slated for Q127 availability and utilizes a bidirectional (BiDi) 2-wavelength (1311/1331nm) multiplexing architecture. The module delivers 6.4Tbps per direction (12.8Tbps aggregate) via 32 ports at 200 Gbps per lane. By migrating from unidirectional to BiDi, a typical 512 XPU scale-up cluster (32Tbps/XPU) saves 240km of fiber, eliminates 20,000 connectors and potential failure joints, and yields 16% reduction in total network spend.
- **Laser scaling and supply chain implications.** Lightmatter's Very Large Scale Photonics (VLSP) laser solution, featuring 128 lasers on a single chip built at TSMC, offering 0.1 FIT rate and capable of supporting 64 colors. Lightmatter's roadmap for SiPh projects NPO adoption in 2027, CPO in 2028, and interposer integration beyond 2029. Lightmatter also has eCLICK and vCLICK detachable solutions with minimal insertion loss.

Heterogeneous Integration Global Summit

"Heterogeneous integration for AI era"

Speaker: Dr. James Chen, TSMC

- **AI performance scaling continues to drive heterogeneous integration evolution.** The speaker highlighted that AI system scaling increasingly depends on heterogeneous integration, with progress required across compute performance, memory hierarchy, interconnect bandwidth, power delivery, and cooling. The company's 3DFabric platform integrates advanced logic, SoIC, HBM, silicon photonics, and CoWoS technologies as a unified architecture to support next-generation AI systems.
- **System and Technology Co-Optimization (STCO) is pivotal for next-generation AI systems.** Beyond transistor scaling, the speaker emphasized coordinated optimization of package architecture, power delivery networks, thermal management, and advanced packaging technologies. The company outlined innovations including SoIC scaling, embedded power delivery components, integrated voltage regulation, and advanced cooling solutions to address increasing AI package complexity and power density.
- **Eco-system collaboration is essential to enable innovations in design, architecture, materials, and tools.** TSMC continues to expand industry collaboration through its 3DFabric Alliance and the development of the 3Dblox modular design language. These initiatives aim to improve interoperability across the semiconductor ecosystem, streamline multi-die design flows, and accelerate adoption of AI-driven design methodologies.
- **Silicon photonics and COUPE extend future AI scaling beyond electrical interconnects.** TSMC showcased COUPE as a tightly integrated photonic engine combining electrical and photonic ICs to address growing I/O bandwidth requirements. The company views co-packaged optics as an important path for improving communication efficiency and reducing latency as AI clusters continue to scale up and scale out.

"Scaling the agentic AI era with advanced packaging, heterogeneous integration, and optical interconnects"

Speaker: Dr. Raja Swaminathan, AMD

- **AI-driven compute shift.** AMD argued that AI is entering an agentic era where infrastructure must support continuous inference, reasoning and tool use rather than traditional chatbot workloads. As AI models scale and inference demand accelerates, system architectures must move data more efficiently across CPUs, GPUs, memory and networking. The company highlighted a growing need for balanced CPU-GPU infrastructure to support agent-based workloads.

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- **System-level optimization for energy efficiency.** AMD emphasized that future datacenter efficiency will be driven by locality, bandwidth utilization and system architecture rather than simply adding more compute. The company sees datacenter energy consumption and data movement as key bottlenecks, making tighter integration across the stack increasingly important. AMD highlighted the industry's transition from device-level optimization toward system technology co-optimization (STCO).
- **HELIOS and rack-scale AI architecture.** AMD positioned HELIOS as a purpose-built rack-scale AI infrastructure platform that integrates compute, memory, networking and cooling into a unified system. The architecture combines MI455X GPUs, EPYC CPUs, scale-up fabrics and scale-out networking to optimize AI factory deployments. AMD stressed that packaging, tray, switch and rack design must be co-developed to maximize overall system efficiency.
- **Heterogeneous integration as the next scaling path.** AMD highlighted advanced packaging, 3D hybrid bonding and high-density memory integration as key enablers of future AI performance scaling. The MI455X platform extends this approach through chiplet-based design, large-scale HBM4 integration and advanced CoWoS-L packaging. Looking ahead, AMD sees tighter compute-memory integration, co-packaged optics, panel-based packaging and next-generation thermal solutions as critical technologies for future AI systems.

"Scaling AI Infrastructure: The Role of Heterogeneous Integration in Co-Packaged Optics"

Speaker: Dr. Vladimir Stojanovic, Ayar Labs

- **Optical scale-up fabrics are becoming the key lever to improve both AI interactivity and throughput.** The speaker argued that AI system performance is increasingly constrained by connectivity rather than compute, with future gains dependent on scaling fabric bandwidth, memory bandwidth and memory capacity. Its TeraPHY™ optical engines are positioned to enable co-packaged optics (CPO) scale-up architectures and extended memory configurations that improve both tokens-per-second-per-user and throughput-per-TCO.
- **Co-packaged optics enables AI clusters to scale beyond rack-level limitations.**The speaker highlighted that traditional electrical I/O faces reach and signal integrity constraints, creating power and density challenges as GPU counts increase. By replacing copper interconnects with optical scale-up fabrics, CPO architectures can extend connectivity across dozens of racks while maintaining lower energy-per-bit and supporting larger GPU domains. Ayar Labs demonstrated an optically connected AI rack ecosystem with partners including Wiwynn, GUC, Corning, SENKO and FOCI.
- **TeraPHY optical engines are designed for high-bandwidth accelerator connectivity.** Ayar Labs showcased a future AI accelerator architecture integrating eight TeraPHY™ optical engines and eight HBM stacks on a common substrate. The platform offers over 50Tbps of scale-up bandwidth per accelerator, supports hundreds of scale-up links, and a micro-ring multi-wavelength architecture designed for high-volume manufacturing.
- **Hybrid bonding is Ayar Labs' preferred EIC-PIC integration approach for future optical engines.** Ayar Labs compared fan-out, 3DIC, hybrid bond and monolithic EPIC approaches across bandwidth, energy efficiency, area efficiency and manufacturability metrics. The speaker highlighted hybrid-bonded EIC/PIC integration as providing the strongest balance of bandwidth density, advanced-node EIC enablement and high-volume manufacturability. This architecture also allows flexible die-to-die electrical interfaces and top-side optical access for detachable optical connectors, simplifying assembly, testing and deployment.
- **An open ecosystem is forming around optical scale-up infrastructure for AI.** Ayar Labs emphasized collaboration across the value chain, including ASIC design partners such as Alchip, MediaTek and GUC, photonics manufacturing through TSMC, assembly and test via TSMC VCA partners, and ODM participation from Wiwynn. Ayar Labs believes this ecosystem approach is necessary to commercialize optical scale-up networks and enable larger AI clusters with stronger token economics.

"Enabling the next era of heterogeneous integration with panel-level packaging innovation and commercialization"

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Speaker: Mr. Michael Rosshirt, Applied Materials

- **AI scaling is shifting from transistor shrinks to heterogeneous integration and panel-level packaging.** The speaker argued that AI compute demand now requires simultaneous scaling of I/O density, power efficiency, form factor and bandwidth, making packaging architecture the key enabler for future HPC and AI systems. Heterogeneous integration, 2.5D packaging and HBM are extending the scaling roadmap as larger packages integrate more silicon, memory, and interconnects.
- **Applied Materials' end-to-end panel equipment portfolio is designed to enable next-generation AI packaging architectures.** AMAT's portfolio combines Digital Lithography Tool (DLT), panel dry etch, CVD, PVD, NEXX P500 electroplating and metrology solutions to address key panel-level packaging challenges across patterning, deposition and interconnect formation. The speaker emphasized that panel technology leverages both front-end semiconductor precision and large-area display manufacturing expertise to support packaging at 510 x 515mm panel scale.
- **Materials innovation is positioned as a key differentiator in panel-level packaging.** The speaker highlighted tunable CVD film properties including stress, refractive index, modulus and adhesion, enabling support for both RDL and high-aspect-ratio TGV applications. He also emphasized low-temperature dry etch processing below 150°C, high-aspect-ratio through-hole plating, and materials optimized for electrical and mechanical performance at panel scale.

FOPLP Innovation Forum

"310mm panel-level platform for next generation package solutions"

Speaker: Mr. Teck Lee, ASE

- **Next generation package architecture.** ASE positioned future AI/HPC package architectures around four key building blocks: high-performance compute integration (FOCoS), advanced interposer carriers, power delivery, and high-bandwidth connectivity through SiPh/CPO. As compute density and memory integration increase, packaging is becoming a system-level enabler rather than simply an interconnect layer.
- **Interposer trend toward larger reticle sizes.** ASE highlighted that AI/HPC packages are rapidly scaling from ~3.5x reticle designs toward 5.5x, 7.5x and eventually 9.5x reticle sizes, driven by larger ASICs and increased HBM integration. The company expects continued growth in package size and integration complexity as next-generation AI accelerators require larger interposer footprints..
- **Panel productivity advantage.** ASE believes panel-level packaging becomes increasingly attractive as reticle size grows. For reticle sizes above 5.5x, a 300-310mm panel can deliver >1.78x productivity versus a 300mm wafer, with even larger gains possible on future 600-620mm panel formats. The company is using 310mm panels as the initial production platform while maintaining a roadmap toward larger panel sizes.
- **310mm panel readiness for AI/HPC.** ASE demonstrated both FOCoS and FOCoS-Bridge implementations on 310x310mm2 panels, including designs with reticle sizes above 3.0x and RDL structures with up to 8 layers. Results showed improved output relative to wafer-based manufacturing while maintaining good RDL topography, via quality and process control. ASE stated its panel-level manufacturing platform is ready to support next-generation AI/HPC packaging requirements.
- **Wafer versus panel process benchmarks.** ASE compared key process differences between wafer and panel manufacturing, focusing on photoresist coating and exposure technologies. Panel processing benefits from slit coating with >90% material utilization and maskless LDI exposure, which provides greater design flexibility and software-based compensation, while traditional wafer processing continues to offer advantages in thin-film uniformity and mature manufacturing infrastructure.

"FOPLP for advanced packaging: promise, practical limits, and the road to scalable manufacturing"

Speaker: Mr. Jonas Hsu, MediaTek

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- **Why industry is moving from wafer to panel.** MediaTek highlighted panel-level packaging as a response to larger AI package sizes, better material utilization, and growing CoWoS capacity constraints. A rectangular panel provides ~3.7x the area of a 300mm wafer and improves utilization as reticle sizes scale, with a 5.5x reticle fitting 16 interposers on panel versus 9 on wafer. The company also cited reticle scaling, capacity bottlenecks, and lower cost structures for large packages as key drivers behind industry migration toward panel formats.
- **Future of chiplets and heterogeneous integration.** MediaTek emphasized that future heterogeneous integration requires ecosystem-wide co-optimization across customers, foundries, OSATs, suppliers and EDA vendors. Key technology trends include reticle scaling, migration from 2.5D to 3D integration, higher power consumption, increasing optical connectivity adoption, and the evolution from standard HBM toward customized memory architectures. The company views vertical integration and design co-optimization as essential to balancing performance, power, thermal, reliability and cost requirements simultaneously.
- **FOPLP technical challenge.** While FOPLP offers meaningful cost-reduction potential, MediaTek identified physical stress, non-standard processes and equipment, and longer debug cycles as the primary challenges. Key technical hurdles include yield management, warpage control, design-process interactions, ecosystem standardization, specialized tooling and extended NPI timelines for complex AI packages. The company noted that successful scaling will require tighter process control and ecosystem development as package sizes and integration complexity increase.
- **Key takeaway.** MediaTek believes panel and wafer formats are complementary rather than directly replacing one another. Wafer-based manufacturing currently benefits from a more mature ecosystem, better warpage control and proven fine-pitch capability, while panel formats offer superior geometric utilization and stronger long-term throughput scaling. Choosing between the two depends on the application's maturity, economics and scaling requirements.

"Enabling system integration in FOPLP: high-precision sub-micron interconnects at panel scale"

Speaker: Dr. Choon Khoon Lim, ASMPT

- **Panel-level system integration.** AI and HPC are driving larger package bodies, greater memory bandwidth, more dies per package and higher interconnect density. FOPLP enables large heterogeneous systems, shorter interconnects and improved scalability, while rectangular panels offer higher theoretical area utilization for large packages. However, the economic advantage depends on maintaining full-panel yield and process uniformity.
- **Improving panel-scale yield.** Yield reflects cumulative variation across die placement, molding, RDL formation, bumping and final inspection. Key challenges include molding-induced die shift, warpage, shrinking RDL line width and spacing, and solder-joint quality. Chip-first flows can use adaptive RDL to compensate for post-placement die shift, while chip-last flows provide a stronger known-good-RDL opportunity but require tighter placement and bonding control.
- **Placement, metrology and adaptive RDL.** Precised die placement is the starting point for panel-level yield, particularly as downstream RDL overlay requirements tighten. Online mapping, metrology and data-driven compensation are needed to manage die shift, local distortion and panel warpage. Panel steppers offer higher throughput, while laser direct imaging provides more flexible localized compensation.
- **Fine-pitch bonding and integrated manufacturing.** Thermo compression bonding (TCB) extends FOPLP into high-density integration by combining heat, force, alignment and interface control. ASMPT's Active Oxide Removal process removes metal oxides before bonding and supports a fluxless TCB flow that eliminates downstream flux cleaning. Its integrated approach combines high-accuracy placement, panel handling, warpage control, inspection and process feedback to support precision manufacturing at scale.

Power and Compound Semiconductor Forum

"Solving AI DC power challenges with advanced packaging"

Speaker: Dr. Harrison Chang, ASE

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- **Packaging expands from chip integration to system optimization.** ASE argued that packaging is no longer limited to electrically connecting chips, but is increasingly responsible for system-level optimization across power delivery, signal integrity and optical interconnects. As AI systems scale, packaging must support heterogeneous integration, active power management and co-packaged optics (CPO) at the system level. Growing AI requirements across compute, memory, package size, power and thermals are driving a broader role for advanced packaging.
- **AI power delivery is becoming a critical packaging challenge.** ASE highlighted that increasing AI compute density is driving substantially higher current requirements, with future high-performance packages requiring embedded power delivery solutions. The company outlined the HPC power conversion chain from datacenter power racks through intermediate converters and voltage regulator modules, positioning packaging as a key enabler of efficient power distribution. ASE sees opportunities to support high-, middle- and low-voltage power conversion through integrated power modules and power SiP technologies.
- **Vertical power delivery improves efficiency and system integration.** ASE showcased the industry's migration from lateral to vertical power delivery architectures, which can reduce path loss and improve overall system efficiency. To support this transition, ASE is developing a toolbox of embedded power technologies including powerSiP, embedded inductors, magnetics-in-substrate solutions and capacitor arrays. The company noted that bringing power delivery components closer to the processor can improve electrical performance, although thermal management becomes increasingly important.
- **VRM and SiP-chiplet integration represent the next packaging frontier**. ASE's VRM roadmap focuses on supporting larger CoWoS-based AI systems through tighter integration of passive and active power components, including future copper-based structures and embedded power solutions. The company expects future packages to incorporate greater levels of power delivery, thermal management and component integration directly within the package. ASE believes its combined OSAT and system integration capabilities position it to support co-design efforts as customers increasingly optimize ASICs, power delivery and system architectures together.

"Accelerating compound semiconductors: innovation and supply chain evolution under AI demand"

Speaker: Mr. Poshun Chiu, Yole Group

- **Accelerated product cadence and system level thermal/CTE co-design.** AI server upgrade cycles have compressed from historic 3-to-5 year horizons down to 6 month intervals, dramatically exacerbating power delivery and thermal dissipation constraints. With greater chip integration driving severe coefficient of thermal expansion (CTE) warpage, legacy isolated packaging interventions are obsolete. The supply chain must also pivot to total system-level cooling architectures—incorporating liquid cold plates, inner manifolds, and even micro-channel lid (MCL). Historically, package-level and system-level engineering operated independently; today, upstream-downstream co-design at the earliest conception phases is a prerequisite to stabilize physical architectures before tapeout.
- **3.5D yield economics, material volatility, and equipment CIP scaling.** New grid-to-chip architectures are evolving from traditional PSU designs toward sidecar power racks, high-voltage IBCs and eventually solid-state transformers (SSTs). Yole highlighted GaN as particularly well-positioned for high-voltage IBC applications enabled by emerging 400V-800V power distribution schemes, while SiC is expected to benefit from SST (Solid State Transformer) adoption due to its high-voltage, efficiency and reliability advantages. The same technologies are also expected to support emerging physical AI applications such as robotics.
- **Photonics is becoming critical for AI connectivity scaling.** As advanced packaging evolves from 2.5D toward high value 3.5D heterogeneous integration, compounding yield loss exerts severe economic penalties on HVM scalability. Simultaneously, rapid material iterations risk stranding capital through equipment modifications and supply chain instability. Equipment vendors must leverage rapid continuous improvement process cycles and broad ecosystem partnerships to

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adapt to shifting specifications while scaling volume output. Furthermore, localized debug and infrastructure support (electricity/water provisioning) remain structural bottlenecks for mid-sized supply chain expansions attempting to aggressively pace TSMC's growth curve.

- **Supply chains are evolving to meet AI demand.** The compound semiconductor ecosystem is responding through capacity repurposing, larger substrate adoption and manufacturing scale-up initiatives. Yole highlighted growing migration of SiC from 6-inch to 8-inch wafers and InP toward 6-inch production, reflecting stronger demand and improving technology maturity. The industry is also moving toward greater cost competitiveness as larger-format manufacturing reaches scale and yields improve.

Advanced Testing Forum

"Keynote: productizing TeraPHY optical engines – challenges & opportunities"
Speaker: Mr. Vivek Khanzodé, Ayar Labs

- **Commercialization roadmap to 2028 and deployable rack-scale systems**. Ayar Labs outlined a clear commercial timeline to transition optical compute interconnects from engineering samples to hyperscale volume: moving from initial partner sampling in 2024 and EVT/photonic process development in 2025 to OCI optical engine fab bring-up in 2026, DVT/PVT rack validation in 2027, and full production volume ramp into AI systems by 2028. To bridge component-level readiness into deployable infrastructure, Ayar Labs partnered with Wiwynn to demonstrate an end-to-end, 100% liquid-cooled reference rack (publicly showcased at OFC and Comutex 2026). The system supports 1,024+ accelerators across racks, delivers 200+ Tbps of optical bandwidth per accelerator, and integrates external ELSFP laser modules to validate field serviceability at cluster scale.

- **Thermal validation of microring modulators and external laser architecture.** Addressing historical industry skepticism regarding the temperature sensitive and resonance drift of microring resonators (MRMs), Ayar Labs presented empirical data showing that closed-loop thermal tuning is an engineering-solved problem rather than a physics barrier. Public data demonstrated stable Bit Error Rates (BER) across 30-80 degrees Celsius thermal cycling in liquid-cooled XPU environments. To preserve overall system reliability, Ayar Labs strictly decouples the laser source from the electronic package. Utilizing hot-swappable external ELSFP module—demonstrated at surface temperatures of around 56 to 57 degrees—prevents burying the primary optical failure mechanism inside high-cost multi-chip packages (MCPs).

- **Three-tier reliability framework, test moat, and standardization.** Ayar Labs framed co-packaged optics productization primarily as a test and supply chain challenge rather than an optical physics issue, emphasizing that customer qualification hinges on proving reliability across a three-tier accountability chain: device level (owned by Ayar Labs), MCP level (owned by XPU customers), and fleet scale (owned by hyperscalers). Commercial deployment criteria mandate a BER of 1e-12 over the long run alongside high temperature, volume qualified optical connectors. To avoid fragmented proprietary architecture, Ayar Labs is actively driving open ecosystem standards—converging on UCIe for electrical die-to-die interfaces, partnering with TSMC for foundry fabrication, and contributing to the JEDEC CPO system qualification task force to establish published reliability benchmarks. The industry needs to share investment in optical-probe capability, with a real step-change in throughput, which is the single biggest lever.

"Testing at scale for the AI era: from silicon complexity to system-level reliability"

Speaker: Mr. Morris D.Y. Chang, KYEC

- **Multi-dimensional complexity and the system-level "good die" redefinition.** The architectural shift from monolithic dies to heterogeneous 2.5D/3D integration dictates that an AI accelerator is no longer a single component, but a highly coupled system encompassing compute, HBM stacks, advanced packaging, high-speed SerDes I/O, power delivery, and firmware. Because complexity is multiplying rather than adding across these domains, test complexity is outpacing transistor scaling. Consequently, traditional pass/fail

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defect screening is obsolete; package integrity, D2D interface robustness, and thermal/power infrastructure (e.g. active load boards and socket cooling) must now be inherently validated as core metrics of baseline silicon quality.

- **The test optimization problem and the AI confidence funnel.** Facing a severe "test time explosion" where total test time equals the sum of all multiplying domains, OSATs are forced to solve a multi-variable optimization problem balancing coverage, reliability margin, test time, and cost. KYEC advocates transitioning from "screening bad" to "Assuring good" via a multi-stage AI Confidence Funnel (screen & detect, margin validation, stress & reliability, and system assurance). This intelligence-driven test allocation aims to maximize reliability confidence per second of test time and per dollar of test cost, effectively lowering the cost per device without compromising field robustness.

- **Fleet reliability, massive scale economics, and full-stack test infrastructure.** At the scale of AI factories, a statistically insignificant 1 DPPM defect rate translates directly into ~1,000 cluster failures, turning rare component anomalies into common operational realities that drive severe business impacts (silent data corruption, node outages, model degradation). To guarantee fleet-level reliability, trust must be built incrementally across the entire test stack: from ATE (wafer/die test) and high-stress burn-in to system level test (SLT) and real-time mission mode validation.

"The state of the art in photonic device testing"

Speaker: Mr. Dave Armstrong, ISE Labs, a subsidiary of ASE, Inc.

- **The 5 insertions of optical test and probing constraints**. For insertion 1 (PIC testing), while baseline platforms like FormFactor's CM300 are proven, the industry is transitioning toward ATE direct-dock solutions (e.g. FormFactor +Advantest+TEL's Triton) and cable-less, high-pin-count interfaces like the Technoprobe Eclipse. Insertion 2 (wafer-level optical engine testing) on platforms like ficonTEC WTL-02 and MPI TS3500DS remains heavily constrained by wafer warpage and the mechanical difficulty of double-sided probing. Consequently, insertion 3 (die-level OE test) is currently perceived as a lower-risk interim path as compared with insertion 2 as die level probing is a proven HVM technology, with vendors like ficonTEC with 3 sites solution and MPI (DTS650-DI) targeting 4-site parallel testing by 2027 to alleviate throughput bottlenecks, ahead of insertion 4 (module level, multiple vendors) and insertion 5 (system level test—focused on mission mode testing).

- **Wafer and package-level burn-in for yield stabilization.** Insertions 1.5 and 4.5 focus heavily on CPO burn-in, which has evolved beyond traditional reliability screening into an essential manufacturing yield and stabilization step. Laser wafer-level burn-in (WLBI) is expanding rapidly because it actively stabilizes sensitive modulators and microrings early in the fabrication process, significantly reducing the downstream scrap rate of highly expensive, fully assembled packages. At the package-level burn-in stage, systems must now manage extreme thermal envelopes, with power dissipations surpassing 2kW and requiring sophisticated multi-zone thermal control architectures to accommodate expanding package footprints.

- **The disaggregated optical rack paradigm for ATE integration.** The structural bottleneck in high-volume CPO testing is not the test handler, but the design of a fungible, production-ready optical equipment rack, as legacy ATE systems lack native optical instrumentation. A viable "optical rack" must integrate seamlessly into ATE software, eliminate manual fiber juggling, provide automated system-level calibration, and support concurrent multi-site testing. Because high-fidelity optical instruments (lasers, polarization controllers, optical switches, power meters) from dozens of vendors are too capital-intensive to dedicate to a single test cell, ISE Labs advocates for a disaggregated, shared-resource architecture. By mirroring data center topologies and routing remote high-performance optical racks across the test floor via optical switches, OSATs can distribute capital costs across multiple testers and scale HVM capabilities without linear capex growth.

"AI-driven shift left: navigating critical test challenges"
Speaker: Mr. Jian Ting Chen, TSMC

- **Heterogeneous 3DIC scaling drivers and the multi-stage inline test flow**.

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Driven by AI semiconductor demand, silicon reticle limits are forcing an aggressive transition toward dense chiplet integration. Because a single microbump interconnect failure can scrap an entire high-value system, TSMC is mandating an inline test methodology to shift defect detection as far left as possible. The resulting testing continuum expands across discrete insertions: RDL wafer acceptance test (RWAT) at the chiplet stage, wafer-level chip probe (WLCP) post-CoW bonding for inline excursion screening to guarantee known good CoW, and assembly test (AST) post-substrate attach (oS)—which validates Die-to-Die DC/AC, Die-to-HBM DC/AC, and HBM MBIST—all prior to downstream final test (FT) and system level test (SLT).

- **Thermal, mechanical warpage, and high-speed contact probing bottlenecks.** As package thermal design power scales beyond 1,000W and power density surges 8x by 2030, traditional wafer-level chip probe (CP) encounters severe physical bottlenecks. Probe chucks exhibit high thermal capacity and limited sensor response, creating a wide delta between chuck temperature and junction temperature, which combined with pitch scaling from 55 microns down to 25 microns, dramatically increases probes needle burnout risk. Mechanically, expanding package dimensions paired with reduced substrate thickness induce extreme warpage during FT; TSMC is actively transitioning from lidless ring assemblies to fully lidded CoWoS designs utilizing high-thermal-conductivity TIM solutions to prevent thermal runaway and TIM melt during test. Simultaneously, interconnect data rates scaling from >200Gbps toward 448G SerDes require strict impedance matching, which can be addressed via coaxial probe needles at FT but remains physically constrained during CP.

- **Shift-left hardware architecture, specialty component screening, and COUPE testing.** To accelerate test coverage curves and optimize test times, TSMC is deploying an ecosystem-wide shift left toolset. Key hardware initiatives include thermally enhanced probe cards providing top-side active heat extraction, active die-embedded probe cards that can execute SLT and mission-mode patterns directly at CP, and advanced chuck cooling envelopes that elevate thermal dissipation from 1x baseline up to >4x for singulated die probing. Furthermore, integrating active components directly into the interposer mandates new screening capabilities for ultra-low pA leakage, high density embedded deep trench capacitors, and TSV-enabled N16 FinFET PMICs with ultra-thin on-wafer inductors. Moving forward, TSMC is applying generative AI to transition test engineering from manual coding to automated debugging, working on known good CoW testing with high pin count probing, while collaborating with partners on novel double-sided prober architectures alongside vibration isolation for optical FAU alignment for COUPE silicon photonics integration.

"HPC/AI device testing: bridging gaps with integrated solutions"

Speaker: Oscar Lee, TSMC

- **Multi-contact interface probing and high-speed signal scaling**. The convergence of 3D heterogeneous architectures (SoIC, CoW, and SiPh) forces test architectures to simultaneously interface with diverse contact topologies—including aluminum pads, copper pads, C4 bumps, fine-pitch microbumps, and optical fiber coupling. To validate interconnect data rates accelerating from 112Gbps to 224Gbps and 448Gbps, probe designs are aggressively scaling down in physical length from Gen 1 (4-6mm, <32Gbps) and Gen 2 (3-4mm, 64-112Gbps) to Gen 3 ultra-short probes (<2mm) featuring impedance-matches probe heads. However, shorter probe pins substantially escalate contact force, necessitating mechanical probe head redesigns to avoid pad/microbump damage. TSMC deploys a distributed testing flow for known good die (KGD), and known good stack (KGS), and AST for assembly interconnects. This is to catch defects before components enter multi-week packaging cycles (>2 weeks for CoW and ~2 weeks for oS), avoiding compounding scrap penalties.

- **Integrated probe card thermal solutions: UCD and TDS.** As compute requirements scale, reticle power delivery at chip probing is increasing by 5x from N5P to N2, with GPU TDP reaching 2,200-2,300W and ASICs exceeding 1,000W. To prevent probe needle burnout under severe electrical loads, uniform current distribution (UCD) technology is introduced—coating metal traces directly onto the probe head to connect the same power rails—which enhances current carrying capacity (CCC) by >40%. Complementing this, the thermal dissipation

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structure (TDS) integrates 3x higher thermal conductivity materials, 5x improved thermal transfer accessories, and a 4x larger tester air purge for heat dissipation. TDS lowers thermal equilibrium temperatures, boosts probe CCC by an additional 20%, and suppresses DUT junction temperature spikes by 3-5 degrees Celsius to avoid thermal runaway. TSMC's active die cooling roadmap scales from air-cooled 25-50W configurations in 2026 to 75W in 2027, moving toward liquid-cooled architectures exceeding 100W by 2028 to enable complex system-level testing (such as direct HBM access) at the wafer probe stage.

- **Active temperature control (ATC) and ultra-high pin count singulated die probing.** To manage intense thermal dynamics across large test surfaces, TSMC is transitioning from legacy single-zone heating and cooling chucks to multi-zone heating and cooling chucks governed by real-time P-I-D (proportional-integral-derivative) control. For singulated die probing, TSMC's showcased integrated solutions that supports massive die dimensions up to 100mm x 100mm and dissipated power densities exceeding 200W/cm^2. The interface accommodates extreme probe card densities surpassing 800k pins while implementing precise mechanical force control to prevent lateral die shifting during high-pressure contact.

3DIC Global Summit

"Panel discussion: key challenges from deployment to scalable manufacturing"

Speaker: Dr. Mike Hung, ASE Inc., Mr. Wen Chin Tseng, C SUN MFG. LTD., Mr. Eric Lin, Jentech Precision Industry Co., Ltd., Mr. William Lin, Wiwynn Corporation, Dr. Jun He, TSMC.

- **Accelerated product cadence and system level thermal/CTE co-design.** AI server upgrade cycles have compressed from historic 3-to-5 year horizons down to 6 month intervals, dramatically exacerbating power delivery and thermal dissipation constraints. With greater chip integration driving severe coefficient of thermal expansion (CTE) warpage, legacy isolated packaging interventions are obsolete. The supply chain must also pivot to total system-level cooling architectures—incorporating liquid cold plates, inner manifolds, and even micro-channel lid (MCL). Historically, package-level and system-level engineering operated independently; today, upstream-downstream co-design at the earliest conception phases is a prerequisite to stabilize physical architectures before tapeout.

- **3.5D yield economics, material volatility, and equipment CIP scaling.** As advanced packaging evolves from 2.5D toward high value 3.5D heterogeneous integration, compounding yield loss exerts severe economic penalties on HVM scalability. Simultaneously, rapid material iterations risk stranding capital through equipment modifications and supply chain instability. Equipment vendors must leverage rapid continuous improvement process cycles and broad ecosystem partnerships to adapt to shifting specifications while scaling volume output. Furthermore, localized debug and infrastructure support (electricity/water provisioning) remain structural bottlenecks for mid-sized supply chain expansions attempting to aggressively pace TSMC's growth curve.

- **Breaking transactional silos via STCO, digital twins, and pre-tapeout engagement.** Late-stage proprietary spec changes by CSPs or fabless customers induce redesigns—such as last-minute thermal specification alteration—has catastrophic impact to component or system suppliers. To resolve this fragmented integration phase for ODMs, TSMC is spearheading a system technology co-optimization (STCO) call-to-action, aiming to implement a fully co-optimized product in the next generation to resolve 50-60% of system verification bottlenecks prior to final assembly. Crucially, digital twins are framed as an absolute imperative for backend advanced packaging; unlike front-end EUV processes which permit rework, backend yield failures are irreversible. Anticipated initiatives like the Baipu pilot line aim to force early-stage collaboration across the localized Taiwanese ecosystem, aligning die/ASIC roadmaps directly with ODM capabilities before CSP designs are locked.

"Keynote: AI-ready equipment: building global competitiveness in advanced packaging equipment"

Speaker: Ms. Lisa Kao, Delta Electronics

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- **Software-driven transformation and physical AI paradigm.** Delta highlighted that competitiveness in advanced packaging equipment has evolved beyond mechanical hardware precision into software-defined, model driven architectures. To build "physical AI" capabilities—enabling equipment to understand, predict, and optimize processes autonomously—Delta establishes a foundation combining precision control (accuracy, stability, determinism) with layered, decoupled software. This creates a trusted context composed of verified ground-truth data and high-fidelity digital twins, allowing equipment makers to scale software complexity without proportional headcount growth through modular platform-based development.

- **SEMI standards evolution and AI-ready architecture (L0 to L5).** Delta mapped its open, modular equipment architecture across the SEMI 2020-2030 standard evolution overlay. The stack spans from baseline hardware (L0 components, L1 mechatronics, L2 real-time machine control) up through L3 machine orchestration, L4 information, to L5 equipment intelligence (closed-loop AI agents and predictive digital twins). Future packaging tools must natively support automation to seamlessly interface with automated fab wide AI orchestration.

- **Case studies to show digital twin validation.** Showcasing practical implementations, Delta presented equipment adaptations for transitioning from round wafer processing (CoWoS, with 50~65% area utilization and sever edge waste) to rectangular panel/square substrate processing (CoPoS, achieving >75% utilization with near zero waste). Leveraging open software frameworks, digital-twin based automated edge path generation, and AI vision guided dynamic compensation, Delta enables round-to-square compatibility across tools including edge profilers, edge grinders, and flatness sorters. Furthermore, Delta demonstrated a "sim-to-real" digital twin pipeline (modifying code on-platform, virtual unit testing and digital safety valuation) to replace legacy, material-intensive trial-and-error commissioning on physical machines, significantly reducing fab move-in lead times and mitigating equipment damage risks.

"Keynote: beyond the reticle: orchestrating the AI hardware ramp"

Speaker: Mr. TY Lau, MediaTek, Inc.

- **AI demand inflection with packaging cost rising**. MediaTek highlighted an unprecedented demand step function, with hyperscaler capex projected to reach around US$1.1T by 2027 (up from US$730bn guided in 2026) and compute requirements doubling every 4 to 5 months (versus the conventional 18–24-month cadence). In this regime, packaging serves as the core enabler: for the B200 accelerator (ex-HBM), advanced packaging, assembly, testing, and yield loss represent 75% of the total estimated unit cost, compared to just 25% for the logic die. As lateral 2.5D scaling is reaching reticle size limits, next-generation performance gains are shifting vertically to 3.5D-on-2.5D architectures—stacking logic dies upward over a silicon interposer flanked by HBM to shorten interconnect lengths, maximize bandwidth, and optimize power efficiency per bit.

- **Compounding yield economics, defect containment, and multi-configuration complexity.** With cluster uptime critically dependent on component reliability, DPPM has elevated into a primary business metric, mandating a strict "contain-own-defect" discipline across every process node. Due to multiplicative yield degradation, a 300bps yield decline at just one manufacturing stage (e.g. dropping from 99.5% to 96.5%) reduces overall integrated four-stage yield from 98.0% to 95.0%; on a 500,000 unit ramp at US $10,000 per unit, this 3% scrap delta translates into around US$150m in unrecoverable losses given >6-month manufacturing lead times. Furthermore, qualifying multi-source permutations across manufacturing sites, HBM stacks, substrates, and raw materials can create up to 24 distinct package build configurations, each carrying unique DFM constraints and requalification overhead.

- **Supply chain village metamorphosis and AI xPU competitive determinants.** Product success in custom AI accelerators is dictated by four pillars: performance per TCO, performance per watt, time to market, and supply assurance. To execute steep ramp curves within the first several months of production, the industry must transition from legacy, sequential vendor

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qualification models to an integrated "supply chain village". MediaTek outlined three essential directives: shift-left on supply chain co-design years ahead of tapeout, establish unified yield intelligence via shared metrology and defect taxonomies to break data silos, and engineer flexible manufacturing capacity capable of absorbing volatile hyperscaler demand signals.

IC Forum-Advanced Chip Technology and Manufacturing

"Keynote: expand AI with leadership silicon & system integration"
Speaker: April Li, TSMC

- **Rapid AI scaling leading to rise with compute demand**. With data center capacity expanding rapidly from 5~6GW/year in 2023 to 30~40GW/year towards 2030, TSMC presents the challenges and opportunities that arise with the "token economics", as it is set to benefit from this megatrend via its leading edge technologies such as CoWoS, SoIC, COUPE, and most importantly, the ability to integrate everything together as a solution.
- **Main challenges that come from AI scaling.** The challenges are: 1) Interconnect bottleneck has risen from a secondary consideration to a primary consideration, as 62% of XPU utilisation is now for data movements, with only 38% of XPU utilisation for compute. 2) Memory wall: memory architecture has not kept the same pace as logic, with memory density also increasing with long context length, multi-modality and multi-user to continue to drive density needs. 3) Power delivery & thermal challenges are escalating with the rise of compute.
- **Opportunities to arise for TSMC from these challenges.** The challenges also bring in opportunities for TSMC. With TSMC's full stack of front end and back end technology platforms, it offers various innovations to solve the bottlenecks. For interconnect, TSMC offers advanced packaging ranging from CoWoS to SoIC and also the COUPE co-packaged optics (CPO) platform. Of note, large form factor 5.5x reticle CoWoS is now in production with >98% yield, with further scaling to come in 2028 (14x reticle and 20 stacks HBM) and 2029 (>14x reticle and 24 stacks HBM). For memory, TSMC can offer Si based technology that enables HBM scaling for HBM bandwidth and efficiency. HBM4 base die with N12 offers >45% power consumption reduction and >2.5x bandwidth. N3 base die will also enhance the next generation HBMs with 2.5x bandwidth improvements.

CEO Summit

"The physics of thought & architecture of intelligence"
Speaker: Amin Vahdat, Google

- **The yield imperative of AI.** The key question is not just what chips or tokens come out of AI infrastructure, but what outcomes come out that improve life. AI is still at the front edge of adoption, and as agentic systems expand into more professions and industries, the focus must increasingly shift from inputs to useful outputs. The industry needs to think about better converting resources into intelligence.
- **Infrastructure innovation to improve output.** The path to true intelligence is not just about bigger models, it is about the radical re-architecture of the compute stack. The AI stack spans energy, data center infrastructure, hardware, software, models and services, creating opportunities for cross-layer optimization. The next frontier comes from architecture and system-level innovation rather than any single component.
- **Network as the nervous system.** The speaker framed Google's networking as a system spanning nanometer chip scale through rack, cluster, planet and even extraterrestrial scale. He highlighted innovations including scale-up ICI networks, optical circuit switching, Falcon hardware transport, and data center network scaling. Networking is increasingly central to delivering efficient movement of data across the AI infrastructure stack.
- **Hardware specialization and efficiency.** Google highlighted a second wave of hardware specialization, supported by TPU supercomputing, specialized silicon features, high-bandwidth interconnect topologies, vertical power delivery and liquid cooling. What makes a TPU is not a single innovation but a system

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architecture including big cores, systolic matrix multiply units, compiler-controlled memory, optical circuit-switched networks and integrated liquid cooling. The speaker emphasized Google is investing heavily in efficiency across its AI stack, from infrastructure and chip design to model deployment.

"AI at scale: rethinking networking, optics and packaging"

Speaker: Asad Khamisy, Broadcom

- **AI networking requirements.** AI is a large distributed computing system with 100K+ XPU clusters working together on a task. As tensor parallelism increases, compute and memory-access latency declines while network latency rises, creating a trade-off between compute speedup and network-latency cost. Desired network characteristics are scalability, resilience, high utilization, and low latency**.**
- **AI scale-up, scale-out and scale-across.** Broadcom positions its portfolio across scale-up, scale-out, and scale-across networking using Ethernet. The portfolio includes Tomahawk Ultra, Tomahawk 6, Jericho, and Thor Ultra NICs, covering HPC, AI scale-up, AI scale-out, and regional scale-out applications. The speaker highlighted Broadcom's complete coverage of HPC and AI with Ethernet.
- **Switch scaling and I/O technologies.** AI makes radix and bandwidth growth valuable, but scaling requires larger die size, package size, interposer size, interconnect size, and thermal solutions. SerDes speed doubles every generation, while increasing lane counts create critical challenges for die configuration and interposer dimensions. Broadcom highlighted co-packaged copper, co-packaged optics, and integrated optics as key I/O packaging paths.
- **Co-packaged optics and advanced packaging.** Co-packaged optics reduce power and improve reliability while supporting future AI interconnect scaling. Key challenges include advanced packaging, optical connectivity, and volume manufacturing, requiring high-density PIC-to-EIC attach and automated assembly. Broadcom also highlighted innovation across mechanical, signal, power-delivery, and thermal technologies to support continued package scaling.

"From silicon innovation to system intelligence: architecting the next wave of AI"

Speaker: Scott J. DeBoer, Micron

- **Memory is foundational for the AI economy.** Memory is no longer secondary to compute and now shapes AI performance, scale and efficiency. Key requirements are bandwidth to feed model performance and high-throughput inference, capacity to expand model context and concurrent workloads, and thermals that determine ultimate usable system performance.
- **Technology leadership enables differentiated solutions.** The speaker highlighted differentiated DRAM and NAND technologies as the foundation for future product leadership. He highlighted leadership in advanced DRAM nodes, DDR5 and LPDDR products, NAND, and data center SSD technologies, with future generations enabling significantly differentiated products.
- **Advanced packaging unleashes the future of memory.** Diverse interconnects, stacking and bonding technologies enable high-performance, low-power memory and logic. Key technology drivers include area and lateral I/O scaling, partitioning, thermals and power, stacking technologies, and interconnects such as micro bumps and hybrid copper bonding.
- **AI infrastructure evolution driven by token-omics.** The data center memory and storage hierarchy spans near memory, main memory, context memory storage and networked data lakes. Memory innovation enables increasing bandwidth through HBM and vertical HBM, while DRAM capacity increases through more stacks per GPU, stack height and Gb per die. Longer term, near-memory compute requires extreme co-design across memory, storage, compute and packaging innovations.

"The next curve: an intelligence imperative"

Speaker: Rani Borkar, Microsoft

- **The yield imperative.** The speaker framed the next phase of AI around "useful

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yield," focusing on maximizing intelligence and outcomes from compute, power and infrastructure. He believes agentic workloads will significantly increase infrastructure demands, making efficiency a key priority**.**

- **Cross-layer optimization drives AI efficiency.** The speaker introduced a yield equation of capability, deployment velocity and utilization. She argued that bottlenecks are rarely solved in one layer, requiring co-design across models, software, networking, memory, silicon and datacenters.
- **Memory and networking as key constraints.** AI workloads are increasingly outpacing memory bandwidth improvements, making data movement and memory efficiency critical. Azure Maia 200 was highlighted as an example of optimizing model architecture, software, silicon and compilers to improve overall system efficiency.
- **Infrastructure co-design extends from datacenter to chip**. The speaker emphasized that power, cooling and silicon design are becoming increasingly interconnected. Azure Cobalt 200 illustrates how workload-aware power optimization can improve efficiency, with future gains expected to come from cross-layer system design.

"The industrialization of intelligence"

Speaker: Michael Kagan, Nvidia

- **Ecosystem of AI models.** The speaker outlined an AI ecosystem that includes both proprietary and open models, with innovation occurring across both tracks rather than converging on a single approach. Management highlighted that proprietary models broaden access to AI, while open models enable customization, tuning, and domain-specific development. Enterprise deployments are increasingly positioned between these two approaches as organizations balance accessibility and flexibility.
- **Era of agentic AI**. The company described a shift from traditional human-computer interaction toward agentic AI systems that can plan, act, execute tasks, and update context through orchestration, tools, memory, and governance layers. This evolution extends AI from knowledge generation toward reasoning and task completion, increasing the amount of useful work produced by AI systems.
- **Digital intelligence enters the physical world.** NVIDIA highlighted the expansion of AI into factories, warehouses, humanoid robots, vehicles, and camera-based systems. Physical AI and simulation technologies are expected to accelerate learning and deployment by enabling large-scale generation of experience beyond real-world limitations.
- **AI factory blueprint**. The company presented AI factories as highly co-designed infrastructure spanning chips, systems, software, networking, storage, and scale-up/scale-out architectures. Management emphasized the importance of optimizing all layers together to maximize performance and resilience at scale. A key addition is NVLink Fusion, which packages the scale-up network architecture as a reusable hardware and software building block that partners can incorporate into their own system designs.

"From chip to data center: how Meta co-designs AI infrastructure"

Speaker: CQ (Chunqiang) Tang, Meta

- **AI infrastructure at scale.** Meta serves 3.6B daily active users and over 200B content recommendations per day. Its AI workloads span LLM training, inference and recommendation systems, requiring different compute, memory and network architectures.
- **Portfolio approach to hardware.** Meta deploys MTIA, NVIDIA and AMD accelerators across AI workloads. MTIA is optimized for GenAI inference, with future generations focused on higher memory bandwidth and performance.
- **Building larger AI clusters**. Meta has deployed hundreds of thousands of MTIA chips and built a 129K GPU cluster by repurposing existing datacenter capacity. The company is also scaling toward gigawatt-class AI clusters.
- **Reliability is critical**. Network failures account for most AI cluster interruptions,

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ahead of GPU and host failures. Meta highlighted silent data corruption and slow GPUs as key risks, as a single faulty component can impact an entire training job.

"Fireside Chat: Technology, Trust, and the Responsibility of Innovation"

Speakers: Tien Wu, ASE, Cliff Hou TSIA/TSMC, Rick Tsai MediaTek, Young Liu Foxconn, S.C. Chien Unimicron

- **AI-driven capacity expansion**. Panelists emphasized that AI demand is scaling rapidly, creating capacity challenges across the semiconductor value chain. The industry is responding with investments in manufacturing, packaging, testing and infrastructure. Ecosystem readiness remains a key bottleneck as demand outpaces traditional planning cycles.
- **Ecosystem collaboration.** AI is driving tighter collaboration across design, manufacturing, packaging and testing. Success increasingly depends on ensuring performance gains at one stage are realized across the entire system. Companies are investing in both capacity and R&D to meet rising performance and energy-efficiency requirements.
- **Taiwan's competitive advantage.** Participants highlighted Taiwan's strengths in semiconductors, packaging and system integration. Flexibility, execution speed and close customer collaboration were cited as key differentiators. These capabilities position Taiwan as a preferred partner for scaling AI infrastructure.
- **Globalization and supply chain resilience.** Speakers described a shift toward a more regionalized manufacturing model driven by customer and geopolitical considerations. Taiwan's role is evolving from a production base to a global technology and manufacturing enabler. Collaboration, knowledge sharing and global resources will be critical to supporting long-term AI growth.

Semiconductor Advanced Inspection and Metrology Forum

"Challenges for CPO testing"

Speaker: Mr. Jeff Lee, Chroma ATE

- **Multi-stage test insertion flow and the "shift to middle" economic pivot**. The co-packaged optics (CPO) test flow spans four primary insertions: Insertion 1 (PIC wafer probe), insertion 2 (EPIC bonded wafer probe), insertion 3 (singulated optical engine/OE test), and insertion 4 (packaged CPO module test). While traditional semiconductor manufacturing prioritizes aggressive "shift-left" strategies, the long test times of wafer-level electro-optical probing (10 to 15 minutes or more per single die at insertion 2) and the mechanical complexity of double-sided probing (top-die optical alignment with bottom-side high-frequency electrical contact) impose highly complex, costly, and elongated test processes. Consequently, Chroma noted that CPO testing could potentially shift toward a "shift-to-middle" paradigm centered on insertion 3. In high-density scale-up switches deploying up to 32 optical engines (e.g. Nvidia's Spectrum-X architectures), any OE defect results in package scrap at the CoWoS level, making 100% known good die (KGD) validation at the singulated OE stage an absolute commercial requirement.
- **Fiber coupling topologies, detachable receptables, and free-space coupling.** Chroma evaluated trade-offs across four prevailing optical coupling architectures: permanent V-groove arrays (lowest insertion loss but difficult to rework), fiber array units (FAUs), detachable receptacles, and Corning GlassBridge. Detachable receptacles have emerged as the dominant mainstream choice (adopted by Nvidia) due to field-serviceability and repeatable passive alignment, despite added interface loss and receptacle unit cost. For insertion 3 OE probing, fine contact pitches preclude conventional mechanical sockets, requiring high-precision vision-guided camera alignment. Furthermore, to mitigate physical wear and collision risks during repeated high-volume docking, equipment vendors are exploring free-space optical coupling to prevent mechanical yield loss during optical engine handling process.
- **Insertion 4 CPO module test, thermal-optical inference, and equipment scalability.** Insertion 4 encompasses conventional electrical final test (Ins4e) and optical continuity verification (ins4o), with high-frequency loopback BER testing

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(Ins4eo) and full system level test (Ins4 SLT) as emerging steps. The primary integration bottleneck is resolving physical mechanical interference between high-power thermal conditioning heads and optical FAU connectors, which mandates decoupled dual-zone temperature control on massive CoWoS substrates susceptible to package warpage. At the system level, SLT requires managing extensive external laser source (ELS) unless embedded on-chip lasers gain adoption. For ATE manufacturers, scaling quarterly production by hundreds of specialized (i.e. around US$1m each) test units create substantial working capital and lead-time pressures, compounded by fragmented customer roadmaps (e.g. disparate micro-ring vs Mach-Zehnder modulator architectures) that limit cross-platform equipment fungibility.

"Keynote: challenges and opportunities for metrology and inspection of advanced packaging"

Speaker: Dr. Zhihua Zou, TSMC

- **Heterogeneous packaging scaling and optical/physical metrology bottlenecks**. As advanced packaging scales toward 1 trillion transistors to power AI compute, package reticles are projected to expand from 5.5x reticle with 12x HBM3E/4 to >14x reticles with 24x HBM5E by 2029, alongside SoIC logic-on-logic pitch shrinking from 6 microns (volume production in 2025) to 4.5 microns at the A14 node by 2029. This structural scaling elevates metrology from a perceived "lean waste overhead" into a critical value-added enabler to enforce zero-defect containment at each packaging stage. Critical dimensions (CD) metrology below 4 micron requires transitioning beyond crowded WLI and confocal solutions toward AFM and OCD targeting <0.5% nominal accuracy, though AFM throughput remains a HVM bottleneck. For 3D bump/ball coplanarity, high-density microbump arrays require chromatic line scan and line confocal systems capable of <0.5% bump height tool-to-tool matching across large die areas, while overlay metrology faces opacity barriers as IR-CCD fails to penetrate non-transparent metals and composite materials like HBM and SiC, so a solution is required.

- **Buried defect inspection and medical imaging cross-pollination.** Detecting sub-surface defects across multi-layer heterogeneous interfaces represents the foremost inspection hurdle in 3D packaging. TSMC is evaluating inspection modalities cross-pollinated from the medical domain (CT, ultrasound, IR, and MRI). While 3D X-ray is effective for detecting solder bridging and bump joint failures, inline implementation faces severe trade-offs between scan speed and resolution, compounded by persistent concerns regarding damage to DRAM/HBM post X-ray scan. Consequently, high-frequency C-mode scanning acoustic microscopy (CSAM >300MHz) is emerging as an alternative to X-ray due to its speed. In parallel, 3D infrared optical coherence tomography (3D IR OCT) is advancing to overcome optical depth-of-focus limits to inspect chipping, cracks, and foreign materials across wafer top, bottom, sides on wafer, frames, dies, and packaging.

- **M&I equipment architecture, warpage management, and design for metrology.** Tool handling and measurement repeatability are challenged by thermal and mechanical warpage across wafer, frame, die, and full-package levels. TSMC outlined a four-pillar aspect for advanced metrology & inspection (M&I) systems: 1) core sensor optics with scalable HVM tool-matching, 2) robot/chuck/handling for high warpage wafer/heavy packaging with strict particle control and self-diagnosis, 3) software/system integration with external systems, and 4) service with localized field support. To overcome physical inspection limits, TSMC called for an industry-wide transition to "design for metrology". This paradigm mandates early ecosystem co-design across fabless designers, foundries, and toolmakers to integrate dedicated overlay/die-shift targets, CD type/film thickness, metrology-friendly material selections, and radiation-tolerant, X-ray-compatible DRAM architectures.

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Neutral	FSR is between -6% and 6% of the MRA.	40%	21%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 June 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

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Company Name	Reuters	12-month rating	Price	Price date
ASE Industrial[16,28]	3711.TW	Buy	NT$588.00	04 Sep 2026
ASPEED Technology Inc[28]	5274.TWO	Buy	NT$17,470	04 Sep 2026
All Ring Tech	6187.TWO	Buy	NT$1,305	04 Sep 2026
Chroma ATE[28]	2360.TW	Buy	NT$2,030	04 Sep 2026
Grand Process Technology[13]	3131.TWO	Buy	NT$2,385	04 Sep 2026
Gudeng Precision Industrial	3680.TWO	Buy	NT$518.00	04 Sep 2026
Hon.Precision, Inc.	7769.TW	Buy	NT$5,980	04 Sep 2026
MediaTek Inc.[28]	2454.TW	Buy	NT$4,415	04 Sep 2026
Taiwan Semiconductor Manufacturing[16,28]	2330.TW	Buy	NT$2,410	04 Sep 2026
UMC[16,28]	2303.TW	Buy	NT$130.00	04 Sep 2026
WIN Semiconductors	3105.TWO	Buy	NT$449.50	04 Sep 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.

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